

Prob 3: In a football match Lionel Messi kicks off a ball into the air with a velocity 25 m/s making an angle of 30° with the ground. After a certain time the ball was received by Neymar at the ground level. Ignore air resistance for all calculations.

- How long does the ball take to reach its maximum height? Also calculate this height.
- Find the velocity (with magnitude and direction) of the ball at $t = 1.5$ sec after the kick off.
- How far was Neymar from Messi?
- Draw the displacement versus time graph for the motion of the ball

Solⁿ (a)

$$v_y = 0$$

$$\Rightarrow v_{0y} - gt = 0$$

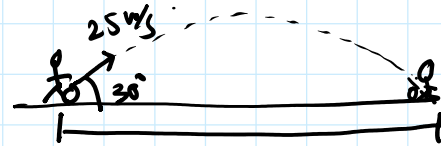
$$\Rightarrow t = \frac{v_y}{g} = \frac{v_0 \sin \theta}{g} = \frac{25 \times \sin 30^\circ}{9.8}$$

$$\Rightarrow \boxed{t = 1.27 \text{ s}} \quad \underline{\text{Ans}}$$

$$y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$$

$$\Rightarrow y = 25 \sin 30^\circ \times 1.27 - 4.9 \times (1.27)^2$$

$$\Rightarrow \boxed{y = 7.97 \text{ m}} \quad \underline{\text{Ans}}$$



$$\underline{\underline{v_y}} = v_{0y} - gt$$

(b)

$$\underline{\underline{v}} = v_x \hat{i} + v_y \hat{j}$$

$$v_x = v_{0x} = v_0 \cos \theta$$

$$v_x = 25 \cos 30^\circ = 21.65 \text{ m/s}$$

$$v_y = v_{0y} + a_y t$$

$$\Rightarrow v_y = 25 \sin 30^\circ - 9.8 \times 1.5$$

$$\Rightarrow v_y = -2.2 \text{ m/s}$$

$$\underline{\underline{v}} = 21.65 \hat{i} - 2.2 \hat{j} \text{ m/s}$$

$$\vec{v} = 21.65\hat{i} - 2.2\hat{j} \text{ m/s}$$

Ans

c)

$$x = v_{0x}t$$

$$\text{or } x = v_{0x} \times 2 \text{ s}$$

$$\text{or } x = 25 \times \cos 31^\circ \times 2 \times 1.2 \text{ s}$$

$$\text{or } \boxed{x = 54.99 \text{ m}} \quad \underline{\text{Ans}}$$

d)

